

LESSON 10 / 第十課

Trees and Construction: From the Project Programme to the Cut Root

樹木與施工：從工程計劃到被切斷的根

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Construction injures trees by four published classes: crown, stem, structural root and absorbing root. Tree protection zones, critical root zones and root protection areas are proxies for where those injuries are likely, not maps of roots. Fite and Smiley's 6–18× DBH guidance, BS 5837's 12× cap and ANSI A300 construction-management language are specification tools for that proxy. Watson and Hauer documented that root loss and grade change have long lag effects. A dripline circle that ignores utilities, slope cuts and the absorbing-root account treats the proxy as a photograph. It is not one.

施工以四種已發表類別傷害樹木：樹冠、莖、結構根與吸收根。樹木保護區、臨界根區與根系保護區是這些傷害可能發生位置的代理，不是根系圖。Fite 與 Smiley 的 6–18 倍胸徑指引、BS 5837 的 12 倍上限，以及 ANSI A300 施工管理語言，是該代理的規格工具。Watson 與 Hauer 記錄了根系喪失與標高改變具有長滯後效應。忽略管線、邊坡開挖與吸收根賬戶的滴水線圓，是把代理當成照片。它不是。

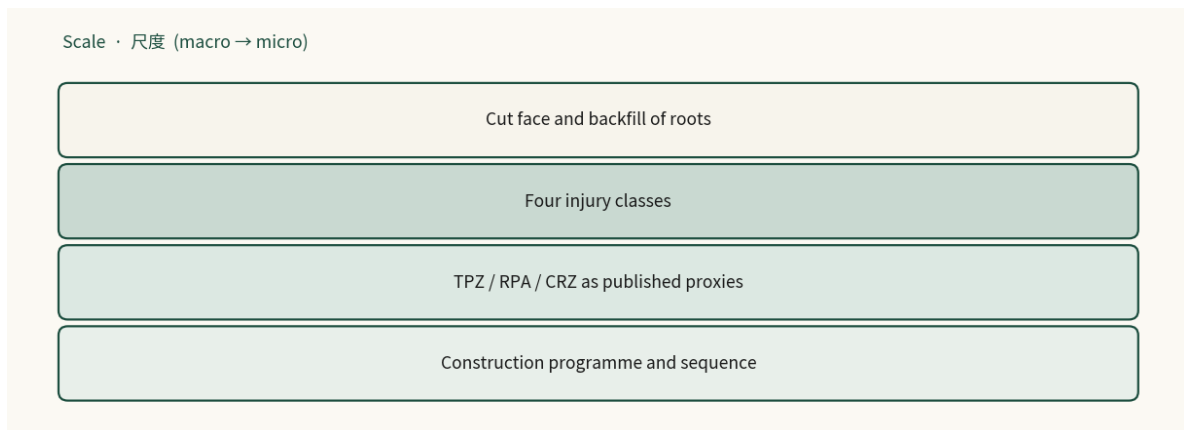


Figure 1. Nested scales from the project programme to the cut root.

圖 1. 由工程計劃到被切根的嵌套尺度。



Figure 2. Published TPZ/RPA rules are proxies with different arithmetic.

圖2. 已發表TPZ/RPA 規則是算術不同的代理。

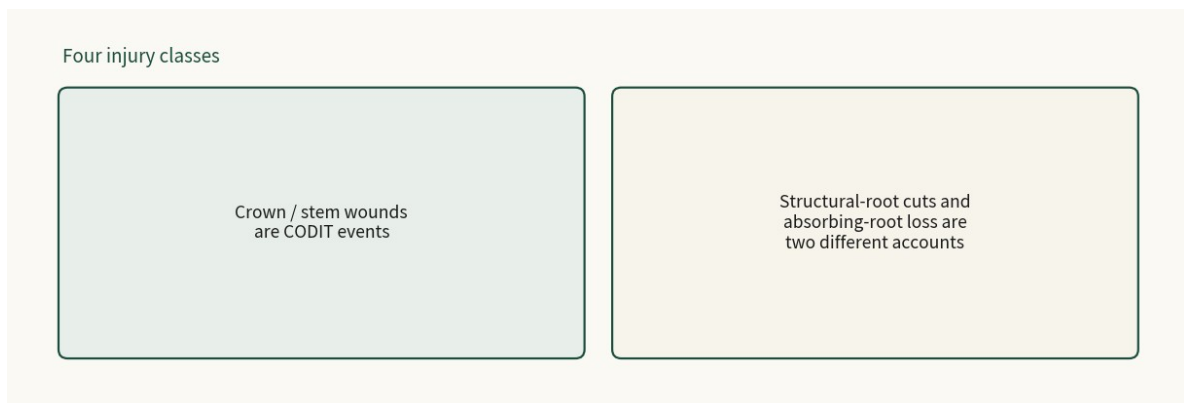


Figure 3. Four injury classes that a single circle does not distinguish.

圖3. 單一圓圈區分不了的四種傷害類別。

1. Scope

1. 範圍

The object is published construction injury and the proxy zones used to manage it. A local ordinance number is an example of a proxy, not the physiology.

對象是已發表的施工傷害，以及用來管理它的代理區。地方法規數字是代理的例子，不是生理。

2. Proxies

2. 代理

Roots do not obey a painted circle. Fite and Smiley reviewed distances commonly used to protect trees during construction and framed 6–18× DBH as a range keyed to tolerance, age and impact, not as a single magic multiple [1]. BS 5837 uses a 12× DBH radius with an area cap [2]. ANSI A300 construction-management language requires a plan, fencing, and a process for work that must enter the zone [3]. Each is a proxy. Treating any one of them as a root map is a category error.

根系不遵守畫出來的圓。Fite 與 Smiley 綜述施工期間常用的保護距離，把 6–18 倍胸徑表述為按耐受力、年齡與衝擊鍵入的範圍，不是單一魔法倍數 [1]。BS 5837 使用 12 倍胸徑半徑並帶面積上限 [2]。ANSI A300 施工管理語言要求計劃、圍欄，以及必須進入該區的工序 [3]。每一個都是代理。把其中任一個當成根系圖，是範疇錯誤。

3. Four injury classes and lag

3. 四種傷害類別與滯後

Crown tearing, stem wounds, structural-root severance and absorbing-root loss are four accounts. The first two are CODIT events on aerial wood. The third changes the mechanical plate. The fourth changes the water and carbon bills. Watson documented multi-year recovery and non-recovery after root loss [4]. Hauer and colleagues measured construction impact as a survival and condition effect that outlasts the contractor's presence [5]. Grade fill over the plate is a chronic oxygen event, not a finished surface.

樹冠撕裂、莖傷、結構根切斷與吸收根喪失是四本賬。前兩項是地上木材的 CODIT 事件。第三項改變力學盤。第四項改變水與碳帳單。Watson 記錄了根系喪失後多年的恢復與不恢復 [4]。Hauer 及同事把施工衝擊測成比承包商在場更持久的存活與狀況效應 [5]。在盤上填土是慢性氧事件，不是完工表面。

4. Limits

4. 限度

Proxy multiples calibrated on temperate open-grown trunks transfer poorly to multi-stem tropical figs, palms, and trees already entangled with walls or utilities. Where the root map is known, the map beats the proxy.

在溫帶開敞生長樹幹上率定的代理倍數，難以轉移到多幹熱帶榕、棕櫚，以及已與牆或管線纏結的樹。凡根系圖已知之處，圖勝於代理。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A TPZ / RPA circle is treated here as:

1. 此處把 TPZ/RPA 圓視為：

- A. A photograph of roots.
- A. 根系照片。
- B. A published proxy, not a root map.
- B. 已發表代理，不是根系圖。
- C. A P50.
- C. 一個 P50。
- D. Wall 4.
- D. 第 4 壁。

2. Fite–Smiley 6–18 x DBH is:

2. Fite–Smiley 6–18 倍胸徑是：

- A. A single magic multiple.
- A. 單一魔法倍數。
- B. A range keyed to tolerance, age and impact.
- B. 按耐受力、年齡與衝擊鍵入的範圍。
- C. A lightning path.

C. 雷電路徑。

D. A diversity ceiling.

D. 多樣性上限。

3. Structural-root cuts and absorbing-root loss:

3. 結構根切斷與吸收根喪失：

A. Are the same account.

A. 是同一賬戶。

B. Are two different accounts.

B. 是兩本不同賬。

C. Both equal t/R .

C. 都等於 t/R 。

D. Both are 3-30-300.

D. 都是 3-30-300。

4. Grade fill over the plate is:

4. 在盤上填土是：

A. A finished surface only.

A. 只是完工表面。

B. A chronic oxygen event.

B. 慢性氧事件。

C. A register listing.

C. 名錄登錄。

D. An inoculant.

D. 接種劑。

5. Watson / Hauer lag effects mean construction impact:

5. Watson／Hauer 滯後效應意味施工衝擊：

A. Ends when the fence comes down.

A. 圍欄拆除即結束。

B. Can outlast the contractor's presence.

B. 可以比承包商在場更持久。

C. Is always zero.

C. 總是零。

D. Equals flash density.

D. 等於閃擊密度。

6. ANSI A300 construction language requires:

6. ANSI A300 施工語言要求：

A. Only a photograph.

A. 只有照片。

B. A plan, fencing, and a process for entry.

B. 計劃、圍欄，以及進入工序。

C. A sacred month.

C. 神聖月份。

D. A palm cut.

D. 棕櫚切。

7. Where a root map is known, the review treats:

7. 凡根系圖已知，本綜述把：

- A. The proxy as sovereign.
A. 代理為主權。
- B. The map as beating the proxy.
B. 圖勝於代理。
- C. t/R as sovereign.
C. t/R 為主權。
- D. Paint as sovereign.
D. 漆為主權。

8. Crown tearing during construction is:

8. 施工期間的樹冠撕裂是：

- A. Free.
A. 免費。
- B. A CODIT event.
B. CODIT 事件。
- C. A 10-20-30 rule.
C. 10-20-30 規則。
- D. A QTRA target.
D. QTRA 目標。

9. BS 5837 12 x is:

9. BS 5837 的 12 倍是：

- A. A photograph of tropical fig roots.
A. 熱帶榕根照片。
- B. A specified proxy with an area cap.
B. 帶面積上限的指定代理。
- C. P88.
C. P88。
- D. Wall 1.
D. 第 1 壁。

10. Four injury classes matter because:

10. 四種傷害類別之所以重要，是因為：

- A. A single percent covers them.
A. 單一百分比就能覆蓋。
- B. They tax different physiological and mechanical accounts.
B. 它們抽不同生理與力學賬戶的稅。
- C. They are paint colours.
C. 它們是漆色。
- D. They set SOS.
D. 它們設定 SOS。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Fite K., Smiley E.T. Best Management Practices: Managing Trees During Construction. ISA.
- [2] BS 5837. Trees in relation to design, demolition and construction.
- [3] ANSI A300 construction-management clauses.
- [4] Watson G.W. (1998 and related). tree-root response after severance.
- [5] Hauer R.J., Miller R.W., Ouimet D.M. (1994). construction impacts on urban trees.
- [6] Matheny N., Clark J.R. Trees and Development.
- [7] Costello L.R., Jones K.S. reducing infrastructure and tree conflicts.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。